~~Decay Ne~~

Artificial Leaky Integrate and Fire Neural Network (ALIFNN)

A bunch of design I came up with for artificial ~~neu~~ nn that try to mimic the biological counterpart without being too computationally intensive. I am listing all the iteration ~~for~~ ~~both~~ so that you understand my thinking process

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Setup :-

Given an input I_t at time step t where I_t is a $n \times 1$ vector, you have to predict the scalar θ_t

Let $I_t = (I_{t,1}, I_{t,2}, \ldots I_{t,n})^T$ where $I_{t,i} \in \mathbb{R} \; \forall i$

W is a $n \times 1$ vector where $W = (W_1, W_2 \ldots)^T$

$\sigma(x)$ = Some activation function

Let P_t = Prediction at time step t

B is a scalar and our bias

Iteration 1:-

Biological neuron automatically store memory in the sense that after a neuron fires, the fired state affects the future firing. The higher more recent ~~the firing~~, the higher the effect firing has a higher effect on the future firing

$$\therefore P_0 = \sigma(W^T I_0 + B)$$

$$\cancel{P_t = \sigma(W^T I_t + B + \alpha \sum_{i=0}^{t-1} P_i h(i; \lambda))}$$

~~h is our decay function and λ controls the rate of decay and is a learned~~

$$P_t = \sigma(W^T I_t + B + \alpha \underbrace{\sum_{i=0}^{t-1} h(t-i; \lambda) P_i}_{z_t(\lambda)})$$

h is our decay function and λ controls the rate of decay and is a learned param

$h(n) \downarrow$ as $n \uparrow$

$\Rightarrow h(t-i) \downarrow$ as $i \uparrow$

$\Rightarrow$ Older prediction have less influence than newer ones prediction

Also let $z_t(\lambda) = \sum_{i=1}^{t-1} h(t-i; \lambda)$

$h(n) \geq 0 \ \forall \ n$ | ~~Also let $z_t(\lambda) = \sum_{i=0}^{t-1} h(t-i; \lambda) P_i$~~

α is a learned scalar. It controls whether a more active past i.e a more +ve z_t ~~+ve $z_t(\lambda)$~~ inhibits the neuron i.e makes the output smaller (happens for -ve α) or does it make the neuron more active i.e makes the output larger (happens for +ve α). It is independent of past for $\alpha = 0$

Iteration 2:-

The ~~dem~~ activation of dendrite of biological neuron is naturally damped i.e it cant get to high or too low (Atleast based on my understanding). So based on that we write:-

$$\cancel{P_0 = \sigma\left(\sum_{i=1}^{n} g W_i P_{t,i} + B\right)}$$

$$P_0 = \sigma\left(\sum_{k=1}^{n} g(W_k P_{0,k}) + B\right)$$

$$P_t = \sigma\left(\sum_{k=1}^{n} g(W_k P_{t,k}) + B + \alpha z(\lambda)\right)$$

g is some function that ~~ha~~ makes sure ~~$W_i P_{t,i}$~~ stays $g(x)$ is bounded $\forall x \in \mathbb{R}$

Note:- g may rely on few learned parameter to determine the range of the output, function shape, etc

Iteration 3:-

The activation of dendrite of biological neuron is also dependent on past activation of the dendrite

Let $A_{t,i} = g(W_i P_{t,i})$

$$\cancel{P_0 = \sigma\left(\sum_{k=1}^{n} A_{0,i} + B\right)}$$

$$\cancel{P_t = \sigma\left(\sum_{k=1}^{n} (A_{t,i} + \alpha_1\right.}$$

~~Same purpose as α_2~~

Let $A_{t,i} = \cancel{g(W_i P}\ g(W_i P_{t,i})$

$P_0 = \sigma\left(\sum_{k=1}^{n} A_{0,k} + B\right)$

$$P_t = \sigma\left(\sum_{k=1}^{n} g\left(A_{t,k} + \alpha_1 \sum_{i=0}^{t-1} h(t-i;\lambda_1) A_{i,k}\right) + B + \alpha_2 z(\lambda_2)\right)$$

α_1 has the same reasoning as α_2

Iteration t:-

$$P_t = \sigma\left(\sum_{k=1}^{n} g\left(W_k P_{t,k} + \alpha_1 \sum_{i=1}^{t-1} h(t-i;\lambda_1) A_{i,k}\right) + B + \alpha_2 z_t(\lambda_2)\right)$$

α_1 has same reasoning and purpose as α_2

Iteration 4:-

For computational simplicity, you ~~would~~ may want $g(x) = x$ as your function and it is not bounded but should work

You may also use $g(x) = \begin{cases} \beta_1, & x > \beta_1 \\ x, & \beta_2 \le x \le \beta_1 \\ \beta_2, & x < \beta_2 \end{cases}$

Ofc, you can use other function. I just took it for simplicity

For $h(x;\lambda) = \exp(-\lambda x)$ is a good choice but you can use others as well.

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